

VAL-D'OR, QC, Canada

WMO: 717250

Lat: 48.0564N

Lon: 77.7867W

Elev: 337

StdP: 97.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-29.2	-36.8	0.1	-32.3	-33.7	0.2	-29.1	10.0	-9.6	9.2	-9.4	2.0	290	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.8	19.6	27.2	18.6	25.5	17.8	21.4	26.1	20.3	24.5	19.3	23.3	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.2	23.3	18.8	14.2	22.4	17.7	13.2	21.4	63.9	26.4	60.0	24.4	56.4	23.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.2	8.2	7.3	DB	-36.7	32.1	3.0	1.5	-38.9	33.2	-40.6	34.1	-42.3	35.0	-44.5	36.1	(4)
(5)			WB	-36.8	23.4	3.0	1.3	-39.0	24.3	-40.7	25.1	-42.4	25.8	-44.5	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-15.8	-13.8	-7.3	1.1	9.8	15.6	18.0	16.4	12.3	5.2	-2.7	-10.6	(6)	
	DBStd	13.12	8.10	7.24	7.34	5.61	5.26	4.50	3.35	3.52	4.61	4.86	5.94	7.31	(7)	
	HDD10.0	3566	799	668	538	271	70	7	0	1	27	164	382	639	(8)	
	HDD18.3	5900	1058	901	796	518	269	106	48	81	189	407	631	898	(9)	
	CDD10.0	802	0	0	1	3	63	174	248	201	96	16	1	0	(10)	
	CDD18.3	93	0	0	0	0	4	23	37	22	7	0	0	0	(11)	
	CDH23.3	910	0	0	1	1	68	258	338	185	58	1	0	0	(12)	
	CDH26.7	198	0	0	0	0	16	63	78	33	8	0	0	0	(13)	
(14) Wind	WSAvg	3.5	3.5	3.6	3.8	3.9	3.7	3.4	3.1	3.0	3.3	3.6	3.7	3.5	(14)	
(15) Precipitation	PrecAvg	981	62	53	69	66	76	98	101	104	105	86	87	76	(15)	
	PrecMax	1217	143	112	136	127	152	196	205	225	191	189	172	116	(16)	
	PrecMin	764	24	14	16	29	13	31	33	41	63	32	27	40	(17)	
	PrecStd	118	25	26	28	25	33	33	40	42	32	32	36	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.0	13.6	20.3	28.9	31.0	31.7	29.5	27.8	22.2	13.1	5.6	(19)	
		MCWB	4.9	3.1	9.3	11.9	17.6	20.0	21.3	20.3	19.6	16.1	10.7	4.5	(20)	
	2%	DB	1.4	1.9	8.4	15.8	25.0	28.4	28.7	27.4	25.0	18.5	9.8	2.6	(21)	
		MCWB	1.0	0.7	4.5	8.1	15.2	18.8	19.8	19.3	18.4	13.8	8.2	2.1	(22)	
	5%	DB	-0.8	-0.1	5.6	12.6	22.5	26.4	26.9	25.5	22.6	15.5	7.2	1.0	(23)	
		MCWB	-1.2	-1.0	2.5	7.0	14.1	17.5	18.7	18.4	17.3	12.6	5.9	0.5	(24)	
	10%	DB	-3.2	-2.3	3.0	9.6	19.5	24.2	25.1	23.5	20.3	13.0	5.0	-0.2	(25)	
		MCWB	-3.7	-3.2	0.8	5.0	12.5	16.4	17.9	17.5	16.3	10.6	3.8	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	3.8	10.4	13.7	20.1	22.2	23.2	22.1	21.1	17.0	11.4	4.8	(27)	
		MCDB	4.9	4.5	12.8	19.0	25.6	28.2	28.5	27.2	25.2	20.4	12.2	5.2	(28)	
	2%	WB	1.0	0.7	5.1	9.7	17.1	20.6	21.5	20.7	19.9	14.6	8.5	2.1	(29)	
		MCDB	1.3	1.6	7.5	13.5	22.7	25.9	25.8	25.1	23.3	17.5	9.3	2.5	(30)	
	5%	WB	-1.2	-1.0	2.8	7.4	15.3	19.2	20.2	19.6	18.4	12.9	6.0	0.6	(31)	
		MCDB	-0.8	-0.2	5.0	11.7	19.9	23.5	24.3	23.5	21.7	15.3	7.1	1.0	(32)	
	10%	WB	-3.7	-3.2	0.9	5.4	13.3	17.9	19.1	18.4	16.7	10.9	3.9	-0.6	(33)	
		MCDB	-3.2	-2.3	2.9	9.2	18.2	22.0	23.4	22.1	19.8	12.8	4.8	-0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	11.7	10.7	12.3	12.1	11.1	10.7	10.4	8.0	7.1	8.7	(35)	
		MCDBR	9.7	10.3	12.4	15.7	16.3	14.7	13.5	13.2	13.0	12.3	9.3	7.2	(36)	
	5% WB	MCWBR	9.6	9.5	9.2	9.8	8.3	7.0	6.1	6.5	7.2	8.4	7.9	7.1	(37)	
		MCWBR	10.0	10.3	11.1	13.9	13.6	11.9	11.1	11.0	11.2	11.3	8.7	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.272	0.280	0.308	0.358	0.361	0.381	0.394	0.378	0.362	0.328	0.303	0.260	(40)	
		taud	2.221	2.215	2.221	2.296	2.404	2.378	2.366	2.403	2.424	2.505	2.441	2.338	(41)	
	Ebn at Noon	Ebn at Noon	794	877	907	892	902	882	864	863	843	823	765	773	(42)	
		Edh at Noon	78	98	116	121	113	117	117	108	96	75	64	61	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.28	2.30	3.76	4.66	5.17	5.63	5.37	4.50	3.44	2.00	1.06	0.86	(44)	
	RadStd	RadStd	0.18	0.33	0.38	0.37	0.48	0.37	0.50	0.31	0.36	0.27	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page